

YANG Hao

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EDUCATION

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- Northwestern Polytechnical University (NPU) | Xi'an, China** Aug. 2022 - Jul. 2026 (Expected)
- Program: **Carbon Neutrality and Smart Energy Innovation Class (Elite Program, Top 20% high school graduates by province)**, Honors College
 - Major: Computer Science and Technology (Bachelor of Engineering)
 - Minor: Smart Energy
 - GPA: **3.739/4.1**
- The Chinese University of Hong Kong, Shenzhen | Shenzhen, China** Jul. 2025 - Aug. 2025
- Program: the 2025 SSE, CUHK-Shenzhen Elite Undergraduate Summer Camp
- University of Oxford | Oxford, United Kingdom** Jul. 2024 - Aug. 2024
- Program: Summer School Program, STEM
 - Grade: **A**
- University of British Columbia | Kelowna, Canada** Jul. 2023 - Aug. 2023
- Program: Summer Program, Engineering
 - Grade: **Gold Medal**

RESEARCH EXPERIENCE

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- Research Program on Remote Sensing Super-Resolution** Jul. 2025 - Aug. 2025
Supervised by Prof. Man-On Pun, CUHK(SZ)
- **Outcome:** Wrote a paper named ***Text-RSSR: Text-Conditioned Remote Sensing Super-Resolution with Cross-Modal Learning***, which is expected to be submitted to IEEE Transactions on Geoscience and Remote Sensing.
 - ✓ Introduced a Text-RSSR model which achieved state-of-the-art performance with PSNR metrics of 16.363 2 dB on Alsat-2B dataset, 27.490 4 dB on UC Merced Land Use Dataset, and 27.408 9 dB on AID.
 - ✓ Developed a dual-head network that extracts fine-grained local visual features from images while simultaneously preserving the global textual and semantic representations captured by CLIP.
 - ✓ Introduced an iterative super-resolution framework that progressively enhances low-resolution images by incorporating both textual cues and semantic information, enabling finer detail recovery and improved alignment with textual descriptions.
- Research Program on Remote Sensing Change Detection** Jul. 2024 - Apr. 2025
Supervised by Prof. Zhiyu Jiang, NPU
- **Outcome:** Wrote a paper named ***Multimodal Difference Augmentation Learning for Remote Sensing Change Detection***, which is now under revision by IEEE Transactions on Geoscience and Remote Sensing.
 - ✓ Introduced an MdaCD model which achieved state-of-the-art performance with IoU scores of 84.88% on LEVIR-CD and 73.13% on SYSU-CD.
 - ✓ Employed a CLIP-Guided Masking process to conceal regions that are highly unlikely to change and subsequently generated text prompts.
 - ✓ Developed a Multimodal Fusion and Difference Augmentation process to enhance the discrepancy between visual and textual information, facilitating differential feature learning and improving the accuracy of change detection.

COMPETITIONS

The 2023 Mathematical Contest in Modeling (MCM)

Feb. 2023

Competitor, Modeler | **Awarded: Honorable Mention**

- Background: Regularity Behind Random Puzzles, three tasks to be addressed.
- Completed Tasks 2 and 3, including problem interpretation, algorithm design, code implementation, and contest paper writing.
- Task 2: Employed a Binary Decision Tree model to analyze the features of the given word and predict the distribution of attempts required to guess the word (1 to 6 attempts or failure).
- Task 3:
 - ✓ Adopted the K-means++ algorithm to classify words into two categories: "simple" and "difficult."
 - ✓ Employed a Binary Logistic Regression model to predict the difficulty of words based on the differences in letter frequency between the two categories.

The 2021 ICPC China Shaanxi Provincial Programming Contest

Oct. 2022

Competitor, Modeler | **Awarded: Bronze Medal**

- Background: Responsible for Problems F and G and collaborated with team members to address Problem B. The Bronze Medal was awarded to the team for completing three Problems (B, G, J).
- Problem F: attempted to optimize a brute-force enumeration algorithm but failed to resolve the Problem within the competition time.
- Problem G: utilized mathematical modeling to calculate the maximum water storage using the arithmetic series sum formula, successfully solving the Problem.
- Problem B: participated in the process by transforming the complex constraints into a mathematical model and using a Binary Indexed Tree (BIT) for point updates and range queries.

ADDITIONAL INFORMATION

- Award and Scholarship:
 - ✓ The 2023 Xiaomi Excellence Award
 - ✓ The 2023 First Class Scholarship (University-level)
- **Programming: C++, Python, MATLAB, LaTeX, Markdown**
- Qualification: Certified Software Professional (CSP), scored 345/500 (Top 2% among all participants)
- Language Proficiency:
 - ✓ English: Fluent (IELTS 7.5, GRE Verbal 150/Quant 167)
 - ✓ Mandarin: Native
 - ✓ Cantonese: Fluent